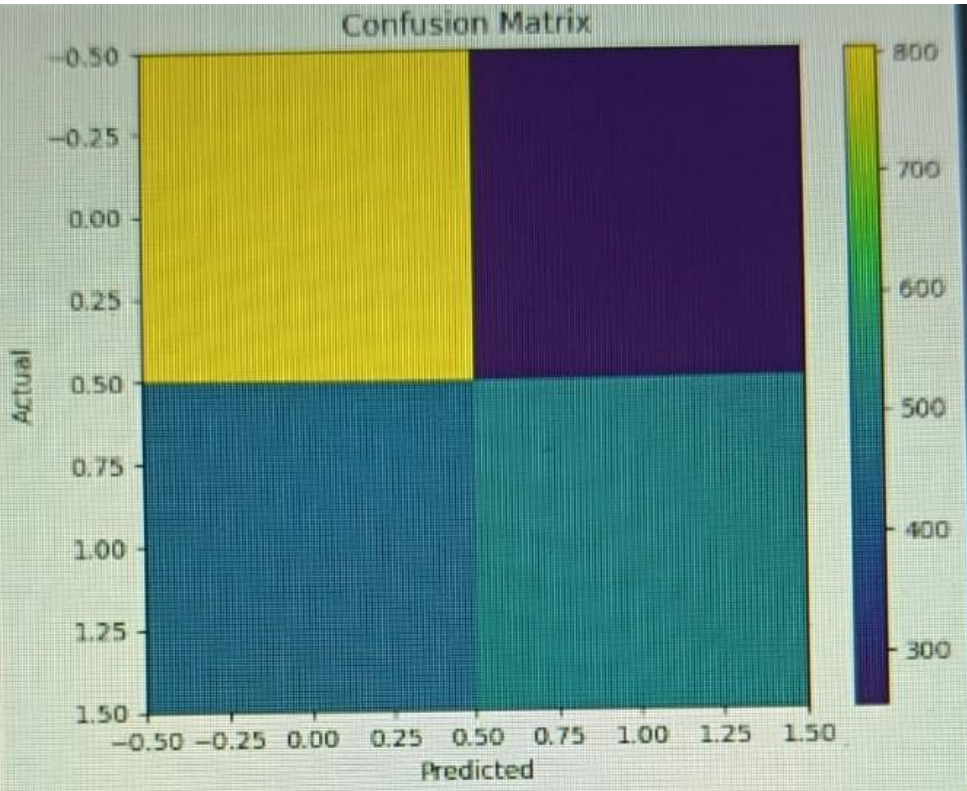
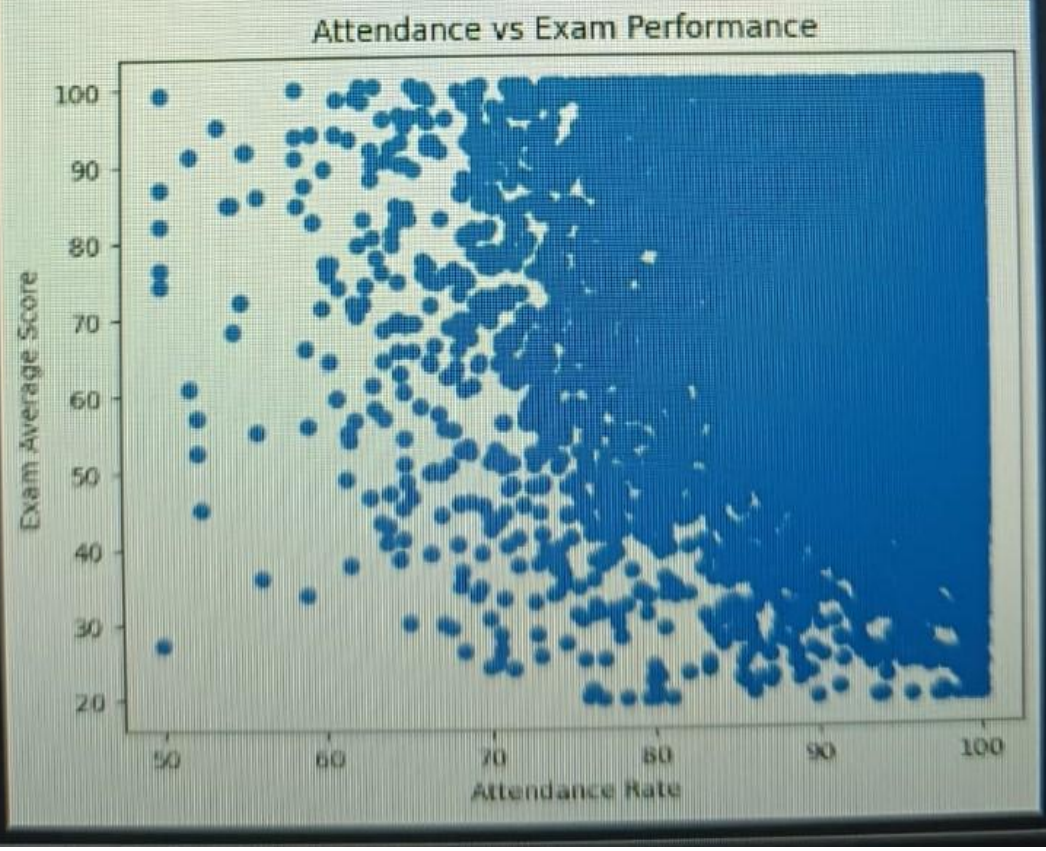
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import precision_score, confusion_matrix

df = pd.read_csv("cleaned_student_dataset-1.csv")

print(df.head())

X = df[[
"attendance_rate",
"assignment_avg_score",
"exam_avg_score",
"homework_submission_rate",
"behavioral_engagement_score",
"digital_learning_hours",
"class_participation",
"extracurricular_participation"
]]

y = df["at_risk"]

X_train, X_test, y_train, y_test = train_test_split(
X, y, test_size=0.2, random_state=42
)

model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)

precision = precision_score(y_test, y_pred)

print("Precision:", precision)

cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:\n", cm)

plt.imshow(cm)
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.colorbar()
plt.show()

plt.scatter(df["attendance_rate"], df["exam_avg_score"])
plt.xlabel("Attendance Rate")
plt.ylabel("Exam Average Score")
plt.title("Attendance vs Exam Performance")
plt.show()
```

```
   age  gender  province  school_type  attendance_rate  assignment_avg_score  \
0    8       0        44            2        89.738773             65.673103
1   18       1        45            1        99.143356             83.875824
2   22       1        44            2        79.106901             47.835205
3    6       1        11            2        94.512831             63.225306
4   21       1        45            0        84.578032             58.824297

   exam_avg_score  class_participation  homework_submission_rate  \
0       65.357657             4.149658                100.000000
1       84.372439             4.645043                 64.595511
2       69.885523             4.486423                 57.884212
3       73.596711             5.945832                 80.236200
4       72.367391             7.136574                 79.240186

   digital_learning_hours  extracurricular_participation  \
0                7.539509                       1.176972
1                4.624189                       4.311543
2                5.589227                       6.183086
3               12.886646                       5.826886
4               12.312776                      10.000000

   behavioral_engagement_score  at_risk  performance_label
0                     8.168861        0                  4
1                     4.518268        1                  0
2                     8.051258        1                  4
3                     7.305437        0                  0
4                     7.949373        1                  0
Precision: 0.6604774535809018
Confusion Matrix:
 [[806 256]
 [440 498]]
```


Confusion Matrix
Actual
Predicted
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
800
700
600
500
400
300


Attendance vs Exam Performance
Exam Average Score
Attendance Rate
100
90
80
70
60
50
40
30
20
50
60
70
80
90
100